



Aakash

Medical | IIT-JEE | Foundations

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FINAL TEST SERIES for NEET-2024

MM : 720

Test - 6

Time : 3 Hrs. 20 Mins.

Topics covered :

- Physics** : Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction
Chemistry : The *p*-Block Elements (Group 15-18), The *d* and *f*-Block Elements, Coordination Compounds
Botany : Principles of Inheritance and Variation, Molecular Basis of Inheritance
Zoology : Reproductive Health, Evolution

Instructions :

- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION-A

1. **Assertion (A):** When radius of a circular loop carrying current is doubled, its magnetic moment does not change.

Reason (R): Magnetic moment is independent of the area of the loop.

- Both Assertion & Reason are true and the reason is the correct explanation of the Assertion
- Both Assertion & Reason are true and the reason is not the correct explanation of the Assertion
- Assertion is true statement but Reason is false
- Both Assertion and Reason are false statements

2. A moving coil galvanometer works on the principle of

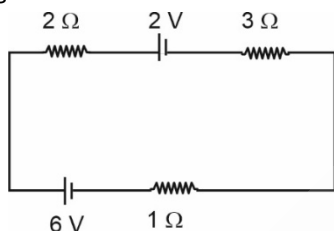
- Chemical effect of current
- Heating effect of current
- Magnetic effect of current
- Conservation of charge

3. A current carrying circular coil has radius R . The distance from the centre of the coil on its axis

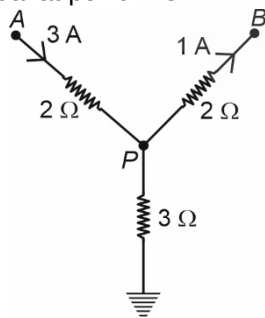
where the magnetic field will be $\left(\frac{1}{27}\right)^{\text{th}}$ of its value at the centre of the coil, is

- $2R$
- $2\sqrt{2}R$
- $2\sqrt{3}R$
- $9R$

4. In an overhead power line current is flowing towards east direction. The direction of magnetic field above the power line is
 (1) Towards west (2) Towards north
 (3) Towards east (4) Towards south
5. If a charged particle is moving in a circle with speed v in a uniform magnetic field with a time period T , then
 (1) $T \propto v$ (2) $T \propto v^0$
 (3) $T \propto v^2$ (4) $T^2 \propto v^3$
6. The current through a wire is dependent on time as $I = \frac{t^3}{3} + 2t^2$, where t is in s and I is in A. The charge flown through a cross-section of the wire in first 3 seconds is
 (1) 24.75 C (2) 29.75 C
 (3) 13.85 C (4) 27.25 C
7. Two batteries and three resistors are connected in a circuit as shown. The current flowing in the circuit is



- (1) 0.83 A (2) 0.33 A
 (3) 1.33 A (4) 0.67 A
8. A part of an electrical circuit is shown in the figure. The potential at point P is

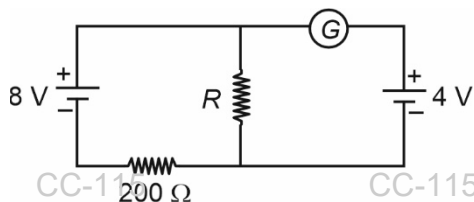


- (1) -6 V (2) 6 V
 (3) 12 V (4) -12 V

9. The work done by a battery of 1.5 V when 6 mC charge is flown through it from its negative to positive terminal
 (1) -9 mJ (2) -4.5 mJ
 (3) 9 mJ (4) 6 mJ
10. Consider the following statements:
 (a) Drift velocity of electrons is much smaller than their thermal velocity.
 (b) Resistance of conductors decrease with increase in their temperature.
 (c) Inside a cell, current always flow from its positive to negative terminal.
 The correct statement(s) is/are
 (1) Only (a) (2) Both (a) and (b)
 (3) Only (c) (4) Both (b) and (c)
11. The unit of intensity of magnetisation is
 (1) A m⁻¹ (2) A m
 (3) T (4) Wb m
12. The torque acting on a magnet of length 5 cm and pole strength 20 A m, kept in a magnetic field of 10⁻³ T at an angle of 30° with field is
 (1) $5\sqrt{3} \times 10^{-4}$ N m (2) 2.5×10^{-4} N m
 (3) 5×10^{-4} N m (4) 10^{-4} N m
13. The correct statement(s) among the following is/are
 (1) An emf can be induced by time varying magnetic field in a coil
 (2) An emf can be induced by moving a circular coil in a variable magnetic field
 (3) In electromagnetic induction, Lenz law violates the conservation of energy principle
 (4) Both (1) and (2)
14. Current passing through an induction coil of inductance 2 H is increasing at the rate of 2.5 A/s. The magnitude of potential difference across the coil is
 (1) 10 V (2) Zero
 (3) 5 V (4) 2.5 V

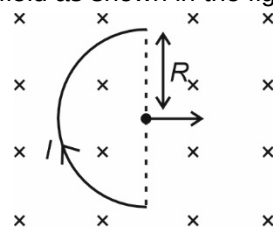
Space for Rough Work

15. Two coils C_1 and C_2 are placed near each other. When current changes by 2 A in coil C_1 , the magnetic flux changes by 0.8 Wb in coil C_2 . The value of mutual inductance of the coils is
 (1) 0.4 H (2) 0.8 H
 (3) 4 H (4) 8 mH
16. If the galvanometer does not show any deflection in the circuit shown, the value of resistance R is



- (1) 100 Ω (2) 200 Ω
 (3) 400 Ω (4) 500 Ω
17. The permeability (in SI unit) of the material of a rod whose magnetic susceptibility is equal to 699, is
 (1) $14\pi \times 10^{-5}$ (2) $20\pi \times 10^{-5}$
 (3) $56\pi \times 10^{-5}$ (4) $28\pi \times 10^{-5}$
18. The magnetic flux linked to a coil is $\phi = (t^3 + 2t + 1)$ Wb, where t is in s. The magnitude of induced emf in the coil at $t = 3$ s is
 (1) 31 V (2) 27 V
 (3) 29 V (4) 25 V
19. The drift velocity of free electron in a conductor is v_0 when the current i is following in it. If both radius and current are halved, the drift velocity will be
 (1) $\frac{v_0}{2}$ (2) $4v_0$
 (3) $2v_0$ (4) $\frac{v_0}{4}$
20. Two wires of same dimension but resistivity ρ and 2ρ are connected in parallel. The equivalent resistivity of the conductor is
 (1) 3ρ (2) $\frac{2\rho}{3}$
 (3) $\frac{4\rho}{3}$ (4) $\frac{\rho}{3}$

21. The dimensional formula of electron mobility inside conductor is
 (1) $[M^{-1}L^2T^{-2}A^{-1}]$ (2) $[M^{-1}LT^2A]$
 (3) $[MLT^{-1}A]$ (4) $[M^{-1}L^0T^2A]$
22. Current density ($\vec{J}$) at an area $\vec{A} = (\hat{i} + 4\hat{j}) \text{ mm}^2$ is $\vec{J} = (2\hat{i} + 3\hat{j}) \text{ A/m}^2$. The current through the area is
 (1) 6 μA (2) 4 μA
 (3) 12 μA (4) 14 μA
23. A charged particle moves through a magnetic field perpendicular to its direction, then
 (1) Both momentum and kinetic energy of the particle are constant
 (2) Kinetic energy changes but momentum is constant
 (3) Momentum changes but kinetic energy is constant
 (4) Both momentum and kinetic energy changes
24. Same current I is flowing in three infinitely long thin wire along positive x, y and z-direction. The magnetic field at a point (0, 0, a) would be
 (1) $\frac{\mu_0 I}{2\pi a}(\hat{i} + \hat{j})$ (2) $\frac{\mu_0 I}{2\pi a}(-\hat{i} - \hat{j})$
 (3) $\frac{\mu_0 I}{2\pi a}(\hat{i} - \hat{j})$ (4) $\frac{\mu_0 I}{2\pi a}(\hat{i} + \hat{k})$
25. The magnitude of magnetic force on a current carrying semi-circular loop moving in a transverse magnetic field as shown in the figure will be



- (1) BIR (2) $2BIR$
 (3) $4BIR$ (4) $\frac{BIR}{2}$

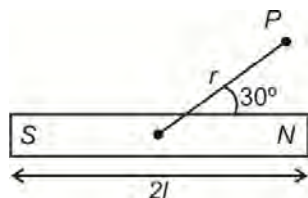
Space for Rough Work

26. The path of a charged particle entering in a uniform magnetic field making an angle 60° with the field is
 (1) Circular (2) Straight
 (3) Elliptical (4) Helical

27. The permanent magnetic moment of the atom of a material is zero. The material

- (1) Must be paramagnetic
 (2) Must be diamagnetic
 (3) Must be ferromagnetic
 (4) May be paramagnetic

28. The magnetic field due to a short bar magnet of magnetic dipole moment M and length $2l$ at a point P which is at a distance r (where $r \gg l$) from the centre of bar magnet as shown in figure is given by



- (1) $\frac{\mu_0 M}{4\pi r^3}$ (2) $\frac{2\mu_0 M}{\pi r^3}$
 (3) $\frac{\sqrt{3}\mu_0 M}{8\pi r^3}$ (4) $\frac{\sqrt{13}}{8} \frac{\mu_0 M}{\pi r^3}$

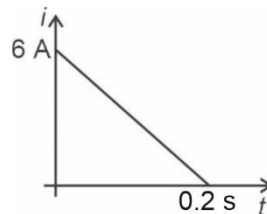
29. If a solenoid has length l , area of cross-section A and number of turns per unit length n , then its self inductance is equal to

- (1) $\mu_0 n^2 A l$ (2) $\frac{\mu_0 n^2 A}{l}$
 (3) $\mu_0 n^2 A^2 l$ (4) $\mu_0 n^2 A l^2$

30. A bar magnet of length $2L$, magnetic moment M and pole strength m is broken perpendicular to its length into two pieces at the middle. The magnetic moment and pole strength of each piece will be

- (1) $M, \frac{m}{2}$ (2) $\frac{M}{2}, m$
 (3) $\frac{M}{2}, \frac{m}{2}$ (4) M, m

31. Magnetic flux is changed through a coil of resistance 5Ω . As a result, an induced current is developed in it, which varies with time as shown in the figure. The magnitude of net change in flux through the coil in Wb is

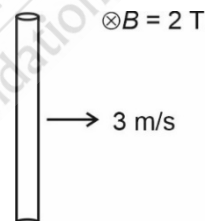


- (1) 2 (2) 3
 (3) 1 (4) 6

32. A metal rod of length 2 m is rotating about its end with an angular velocity of 100 rad/s in a plane perpendicular to a uniform magnetic field of B . If the potential difference between the end of the rod is 60 V. Then magnitude of magnetic field B will be

- (1) 1 T (2) 0.6 T
 (3) 2 T (4) 0.3 T

33. A conducting rod of length 10 cm is moved with 3 m/s in a uniform transverse magnetic field of magnitude 2 T, then the potential difference across its two ends is



- (1) Zero (2) 0.2 V
 (3) 0.6 V (4) 1 V

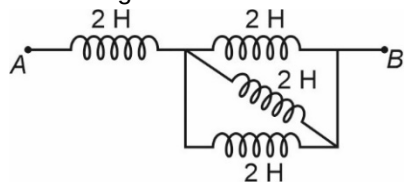
34. Mutual inductance of two coils can be increased by

- (i) Decreasing the number of turns in the coils
 (ii) Increasing the number of turns in the coils
 (iii) Winding the coils on steel core
 Select the correct option.

- (1) (i) and (iii) (2) (ii) and (iii)
 (3) (i), (ii) and (iii) (4) Only (i)

Space for Rough Work

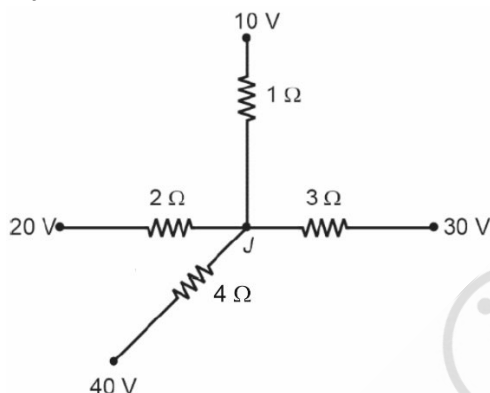
35. The equivalent inductance between points A and B for the given combination of inductors is



- (1) 8 H (2) 6 H
(3) $\frac{4}{3}$ H (4) $\frac{8}{3}$ H

SECTION-B

36. Four resistances are joined together to form a network as shown in the figure. The potential of the junction J will be



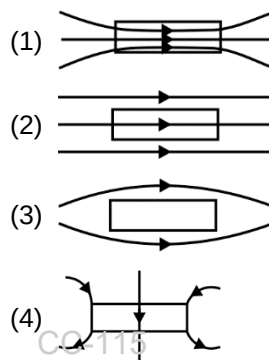
- (1) 19.2 V (2) 17.2 V
(3) 30.2 V (4) 15.2 V
37. A cell develops the same power across two resistance R_1 and R_2 separately. The internal resistance of the cell is

- (1) $R_1 + R_2$ (2) $\frac{R_1 + R_2}{2}$
(3) $\sqrt{R_1 R_2}$ (4) $\frac{\sqrt{R_1 R_2}}{2}$

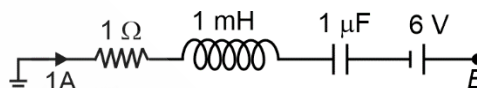
38. A deuteron and an α -particle with same momentum enter perpendicular to uniform magnetic field. Then ratio of radii of their respective circular paths is

- (1) $\sqrt{2} : 1$ (2) 2 : 1
(3) 4 : 1 (4) 1 : 1

39. A superconductor slab is placed in a magnetic field. The correct representation of magnetic field lines in space is

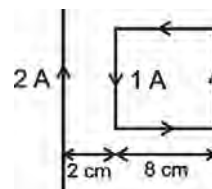


40. If $V_B = 10$ V and instantaneous current 1 A is flowing in circuit which is increasing at the rate of 10^3 A/s as shown in the figure, then charge on capacitor is



- (1) 2 μ C (2) 7 μ C
(3) 6 μ C (4) 10 μ C

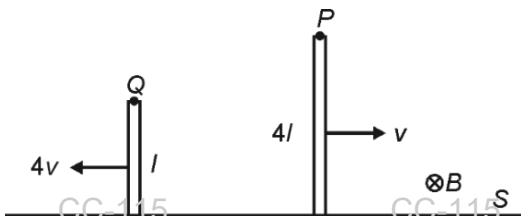
41. A long straight current carrying wire is placed in horizontal plane near a square loop carrying a current of 1 A as shown in figure. The net force on the loop is



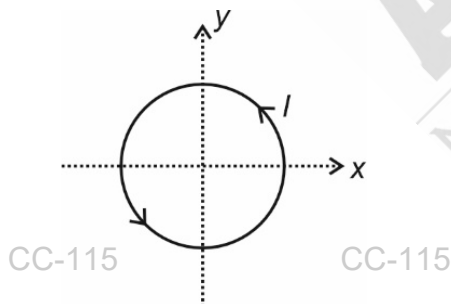
- (1) 1.28×10^{-5} N, attractive
(2) 1.28×10^{-6} N, repulsive
(3) 1.92×10^{-5} N, attractive
(4) Zero

Space for Rough Work

42. Two conducting rods of length l and $4l$ undergo translatory motion in opposite direction with velocity $4v$ and v respectively on a conducting surface S as shown in the figure. There is a uniform magnetic field B into the plane of the paper. The potential difference between the highest points P and Q of the two rods is



- (1) Zero (2) $8lvB$
 (3) $4lvB$ (4) $10lvB$
43. A charged particle enters a uniform magnetic field with velocity vector at an angle of 37° with the magnetic field. The pitch of the helical path followed by the particle is P . The radius of the helix will be
- (1) $\frac{3P}{2\pi}$ (2) $\frac{3P}{8\pi}$
 (3) $2\frac{P}{\pi}$ (4) $\frac{1P}{3\pi}$
44. A circular current carrying loop of radius 5 cm and having 50 turns is placed in $x - y$ plane as shown in figure. A uniform magnetic field $\vec{B} = 2\hat{i}$ tesla exist in the region. If current in the loop is 3 A, then match the following columns and tick the correct option.



	Column-I		Column-II
(A)	Magnitude of magnetic moment (in $A\ m^2$) of the loop	(P)	$\frac{3\pi}{4}$
(B)	Direction of magnetic moment of the loop	(Q)	$\frac{3\pi}{8}$
(C)	Magnitude of torque (in $N\ m$) on the loop	(R)	along $+z$ axis
(D)	Direction of torque on the loop	(S)	along $+y$ axis

- (1) (A) $\rightarrow$ Q, (B) $\rightarrow$ S, (C) $\rightarrow$ P, (D) $\rightarrow$ R
 (2) (A) $\rightarrow$ P, (B) $\rightarrow$ R, (C) $\rightarrow$ Q, (D) $\rightarrow$ S
 (3) (A) $\rightarrow$ P, (B) $\rightarrow$ S, (C) $\rightarrow$ Q, (D) $\rightarrow$ R
 (4) (A) $\rightarrow$ Q, (B) $\rightarrow$ R, (C) $\rightarrow$ P, (D) $\rightarrow$ S
45. A galvanometer with resistance $72\ \Omega$ is changed into an ammeter by using a shunt of $2\ \Omega$. The fraction of total current passing through the galvanometer will be
- (1) $\frac{1}{36}$ (2) $\frac{1}{37}$
 (3) $\frac{2}{37}$ (4) $\frac{3}{71}$
46. Curie temperature is the temperature above which material X becomes material Y. The X and Y respectively are
- (1) Paramagnetic, ferromagnetic
 (2) Paramagnetic, diamagnetic
 (3) Ferromagnetic, diamagnetic
 (4) Ferromagnetic, paramagnetic
47. A battery is connected across metallic conductor of non-uniform cross section. The quantity which remains constant throughout the length of the conductor is
- (1) Current (2) Current density
 (3) Electric field (4) Drift speed

Space for Rough Work

48. A wire of resistance R_0 is stretched by 0.3%, its new resistance will be increased by
- 0.3%
 - 0.6%
 - 0.15%
 - 0.9%
49. In a coil of area 20 cm^2 and 50 turns with magnetic field directed perpendicular to the plane changing at the rate 10^3 T/s . The resistance of the coil is 20Ω . The current in the coil will be

- $\frac{5}{2} \text{ A}$
- 5 A
- 2 A
- $\frac{1}{2} \text{ A}$

50. Two conducting circular loop are concentric and lie in the same plane. The current in the outer loop is anticlockwise and increasing with time, then the induced current in the inner loop is
- Anticlockwise
 - Zero
 - Clockwise
 - First anticlockwise then clockwise

CHEMISTRY

SECTION - A

51. When MnO_2 is fused with KOH and KNO_3 , a coloured compound is formed, the compound is
- K_2MnO_4
 - Mn_2O_7
 - Mn_2O_3
 - Mn_3O_4
52. Metal which is absent in german silver is
- Cu
 - Zn
 - Cr
 - Ni
53. Consider the following statements
- P-P-P bond angle in white phosphorus is 60°
 - Red phosphorus is obtained by heating white phosphorus at 573 K in an inert atmosphere for several days
 - Red phosphorus glows in dark
- The correct statements are
- (a) and (c) only
 - (a) and (b) only
 - (b) and (c) only
 - (a), (b) and (c)
54. Which of the following pair of ions are coloured in aqueous solution?
- Cr^{3+} and Ti^{3+}
 - Sc^{3+} and Ti^{3+}
 - Ni^{2+} and Sc^{3+}
 - Zn^{2+} and Cu^+
55. Hydrolysis of IF_5 forms
- HCl
 - HI
 - HIO_3
 - HIO_4

56. Number of 'P-OH' bonds present in $\text{H}_4\text{P}_2\text{O}_7$ is
- 2
 - 3
 - 1
 - 4

57. The optimum pressure and temperature conditions for the formation of ammonia by Haber's process respectively are
- 200 atm, 70 K
 - 20 atm, 70 K
 - 2000 atm, 700 K
 - 200 atm, 700 K

58. Match the following

	Column-I		Column-II
a.	Silent electric discharge	(i)	H_2SO_4
b.	Ostwald's process	(ii)	Cl_2
c.	Deacon's process	(iii)	O_3
d.	Contact process	(iv)	HNO_3

The correct option is

- a(ii), b(iii), c(i), d(iv)
 - a(i), b(iii), c(iv), d(ii)
 - a(iv), b(iii), c(ii), d(i)
 - a(iii), b(iv), c(ii), d(i)
59. Which of the following statements is correct?
- Among halogens, F_2 is the strongest oxidising agent
 - Negative electron gain enthalpy of oxygen is greater than negative electron gain enthalpy of selenium
 - N_2O is acidic in nature
 - Bond dissociation enthalpy of F_2 is greater than Br_2

Space for Rough Work

60. Which among the following has highest enthalpy of atomization?

- (1) Sc (2) Ti
(3) Mn (4) V

61. Most common oxidation states of copper and titanium respectively are

- (1) +2 and +3 (2) +1 and +3
(3) +2 and +4 (4) +1 and +4

62. Which of the following complexes has tetrahedral geometry?

- (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $[\text{Ni}(\text{CN})_4]^{2-}$
(3) $[\text{Ni}(\text{CO})_4]$ (4) $[\text{PtCl}_4]^{2-}$

63. Correct order of ionic radii of the given ions is

- (1) $\text{Er}^{3+} > \text{Nd}^{3+} > \text{Gd}^{3+}$ (2) $\text{Nd}^{3+} > \text{Er}^{3+} > \text{Gd}^{3+}$
(3) $\text{Gd}^{3+} > \text{Er}^{3+} > \text{Nd}^{3+}$ (4) $\text{Nd}^{3+} > \text{Gd}^{3+} > \text{Er}^{3+}$

64. Correct order of bond angle of the given compounds is

- (1) $\text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S} > \text{H}_2\text{O}$
(2) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$
(3) $\text{H}_2\text{S} > \text{H}_2\text{O} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$
(4) $\text{H}_2\text{Se} > \text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Te}$

65. Given below are the two statements

Statement I: Boiling point of ammonia is greater than phosphine.

Statement II: Ammonia molecules are associated with each other by intermolecular hydrogen bonding.

In light of the above statements, choose the correct answer

- (1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

66. Coordination entity which absorbs shortest wavelength of light is

- (1) $[\text{Cr}(\text{CN})_6]^{3-}$ (2) $[\text{CrF}_6]^{3-}$
(3) $[\text{Cr}(\text{en})_3]^{3+}$ (4) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

67. Match the given compounds in column-I with the hybridisation and shape given in column-II.

	Column-I		Column-II
(i)	XeF_4	(A)	sp^3d^2 – Square pyramidal
(ii)	XeOF_4	(B)	sp^3d – See saw
(iii)	XeO_2F_2	(C)	sp^3 – Pyramidal
(iv)	XeO_3	(D)	sp^3d^2 – Square planar

(1) (i)-(D), (ii)-(B), (iii)-(A), (iv)-(C)

(2) (i)-(D), (ii)-(A), (iii)-(B), (iv)-(C)

(3) (i)-(A), (ii)-(D), (iii)-(B), (iv)-(C)

(4) (i)-(D), (ii)-(C), (iii)-(B), (iv)-(A)

68. Consider the following statements

(i) Copper does not liberate hydrogen on reaction with dilute H_2SO_4

(ii) Metallic radius of titanium is greater than vanadium

(iii) $\text{Ni}(\text{CO})_4$ is an organometallic compound

The correct statements are

- (1) (i) and (ii) only (2) (ii) and (iii) only
(3) (i) and (iii) only (4) (i), (ii) and (iii)

69. White phosphorus on reaction with concentrated NaOH solution in an inert atmosphere of CO_2 gives

- (1) H_2PO_2^- and PO_4^{3-} (2) PH_3 and HPO_3^{2-}
(3) PH_3 and H_2PO_2^- (4) HPO_3^{2-} and PO_4^{3-}

70. Which among the following complex ions is diamagnetic in nature?

- (1) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Mn}(\text{CN})_6]^{3-}$
(3) $[\text{Fe}(\text{CN})_6]^{3-}$ (4) $[\text{CoF}_6]^{3-}$

71. Negative hydration enthalpy of which metal cation is highest?

- (1) Ni^{2+} (2) Ti^{2+}
(3) Cr^{2+} (4) V^{2+}

72. Magnetic moment of Sc^{+x} ion is zero. What will be value of x?

- (1) 1 (2) 2
(3) 3 (4) 4

Space for Rough Work

73. Hydride which has negative enthalpy of formation is
 (1) NH_3 (2) BiH_3
 (3) AsH_3 (4) SbH_3
74. The incorrect match for oxoacids of phosphorus is
 (1) H_3PO_2 : Phosphinic acid
 (2) H_3PO_3 : Phosphonic acid
 (3) $\text{H}_4\text{P}_2\text{O}_6$: Hypophosphorous acid
 (4) $\text{H}_4\text{P}_2\text{O}_5$: Pyrophosphorous acid
75. Which of the following contains S—O—S linkage?
 (1) $\text{H}_2\text{S}_2\text{O}_3$ (2) $\text{H}_2\text{S}_2\text{O}_6$
 (3) $\text{H}_2\text{S}_2\text{O}_7$ (4) $\text{H}_2\text{S}_2\text{O}_4$
76. Consider the following statements
 (a) N_2O_3 is an acid anhydride of HNO_2 .
 (b) Sulphur dioxide reacts with chlorine in the presence of charcoal to give sulphuryl chloride.
 (c) I_2O_5 is used in the estimation of CO.
 Correct statements among the following are
 (1) (a), (b) only (2) (b), (c) only
 (3) (a), (c) only (4) (a), (b) and (c)
77. Metal present in Wilkinson catalyst is
 (1) Co (2) Ni
 (3) Rh (4) Os
78. Crystal field splitting energy, Δ_o will be minimum for
 (1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Co}(\text{en})_3]^{3+}$
 (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
79. XeF_4 upon reaction with SbF_5 gives which ion?
 (1) $[\text{XeF}_6]^{2-}$ (2) $[\text{XeF}_2]^{2+}$
 (3) $[\text{XeF}_5]^-$ (4) $[\text{XeF}_3]^+$
80. Correct order of ligand field strength of the given species is
 (1) $\text{I}^- > \text{F}^- > \text{H}_2\text{O} > \text{NH}_3$ (2) $\text{NH}_3 > \text{H}_2\text{O} > \text{I}^- > \text{F}^-$
 (3) $\text{NH}_3 > \text{H}_2\text{O} > \text{F}^- > \text{I}^-$ (4) $\text{H}_2\text{O} > \text{NH}_3 > \text{I}^- > \text{F}^-$
81. Metal-carbon bond length is maximum in which of the given organometallic species?
 (1) $[\text{Mn}(\text{CO})_6]^+$ (2) $[\text{V}(\text{CO})_6]^{115}$
 (3) $[\text{Ti}(\text{CO})_6]^{2-}$ (4) $[\text{Cr}(\text{CO})_6]$
82. White phosphorus on reaction with thionyl chloride gives
 (1) PCl_5 and SO_2 (2) PCl_5 and SO_3
 (3) PCl_3 , SO_2 and S_2Cl_2 (4) PCl_3 and SO_3
83. The gas evolved on the reaction of platinum with aqua regia is
 (1) O_2 (2) N_2
 (3) NO (4) N_2O
84. Given below are the two statements
Statement I: XeF_4 undergoes disproportionation reaction with water
Statement II: XeF_4 is colourless crystalline solid.
 In light of the above statements, choose the correct answer.
 (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct
85. Which among the following is green in colour?
 (1) CrO_4^{2-} (2) $\text{Cr}_2\text{O}_7^{2-}$
 (3) MnO_4^- (4) MnO_4^{2-}

SECTION - B

86. Given below are two statements, one is labelled as Assertion A and other is labelled as Reason R.
Assertion (A): Ce^{3+} is used as an oxidising agent in volumetric analysis.
Reason (R): Ce^{3+} has noble gas electronic configuration.
 In light of above statements, choose the correct answer.
 (1) Both Assertion & Reason are true and the reason is the correct explanation of the Assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the Assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

Space for Rough Work

87. Correct IUPAC name of $K_3[Fe(CN)_6]$ is
 (1) Potassium hexacyanidoiron (II)
 (2) Potassium hexacyanidoiron (III)
 (3) Potassium hexacyanidoferrate (II)
 (4) Potassium hexacyanidoferrate (III)
88. Correct order of oxidising power of $Cr_2O_7^{2-}$, MnO_4^- and VO_2^+ is
 (1) $VO_2^+ > Cr_2O_7^{2-} > MnO_4^-$
 (2) $MnO_4^- > Cr_2O_7^{2-} > VO_2^+$
 (3) $Cr_2O_7^{2-} > MnO_4^- > VO_2^+$
 (4) $MnO_4^- > VO_2^+ > Cr_2O_7^{2-}$
89. Oxide of nitrogen which has N–O–N bond is
 (1) N_2O_3 (2) N_2O_4
 (3) N_2O_5 (4) N_2O
90. The correct relation between Δ_0 and Δ_t is (Δ is splitting energy gap) represented as
 (1) $\Delta_0 = \frac{4}{9}\Delta_t$ (2) $\Delta_0 = \frac{9}{4}\Delta_t$
 (3) $\Delta_0 = \frac{4}{5}\Delta_t$ (4) $\Delta_0 = \frac{5}{4}\Delta_t$
91. Select the reaction which shows incorrect product formation
 (1) $PCl_5 + H_2O \longrightarrow POCl_3 + 2HCl$
 (2) $POCl_3 + 3H_2O \longrightarrow H_3PO_4 + 3HCl$
 (3) $2Ag + PCl_5 \longrightarrow 2AgCl + PCl_3$
 (4) $C_2H_5OH + PCl_5 \longrightarrow C_2H_6 + POCl_3 + Cl_2$
92. Aqueous solution of 1 mol of $CoCl_3 \cdot 6NH_3$ upon addition with $AgNO_3$ produces 3 moles of white precipitate. Primary and secondary valency of metal in this complex respectively are
 (1) 3, 6 (2) 2, 6
 (3) 3, 3 (4) 6, 4
93. Maximum number of geometrical isomers possible for the square planar coordination complex $[MABCD]$ is
 [M is the central metal atom or ion and A, B, C and D are the monodentate ligands]
 (1) 2 (2) 3
 (3) 4 (4) 5
94. Incorrect statement about Rhombic sulphur is
 (1) It is also known as α -sulphur
 (2) It is of yellow colour
 (3) It is formed by evaporating the solution of roll sulphur in CS_2
 (4) It is soluble in water
95. Element of highest first ionization energy among the following is
 (1) Fe (2) Co
 (3) Ni (4) Zn
96. Which of the following will have lowest molar conductance at infinite dilution?
 (1) $[Cr(NH_3)_4Cl_2]Cl$ (2) $K_3[Fe(C_2O_4)_3]$
 (3) $[Co(NH_3)_5Cl]Cl_2$ (4) $[Cr(NH_3)_5Cl]Cl_3$
97. Consider the following reaction
 $H_3PO_2 + H_2O + AgNO_3 \rightarrow Ag + HNO_3 + X$
 Compound X is
 (1) Metaphosphoric acid
 (2) Hypophosphoric acid
 (3) Orthophosphorous acid
 (4) Orthophosphoric acid
98. Which of the following oxides is basic in nature
 (1) CrO (2) Cr_2O_3
 (3) V_2O_5 (4) Mn_2O_7
99. Match List I with List II
- | | List I | | List II |
|----|---------------------|-------|---------|
| a. | $[Fe(CN)_6]^{3-}$ | (i) | 1.73 BM |
| b. | $[NiCl_4]^{2-}$ | (ii) | 5.92 BM |
| c. | $[Co(NH_3)_6]^{3+}$ | (iii) | 0 BM |
| d. | $[FeF_6]^{3-}$ | (iv) | 2.83 BM |
- Choose the correct answer from the options given below
 (1) a(i), b(iv), c(iii), d(ii) (2) a(i), b(ii), c(iii), d(iv)
 (3) a(iv), b(iii), c(ii), d(i) (4) a(ii), b(i), c(iii), d(iv)
100. The correct electronic configuration of the central metal ion in $[Fe(CN)_6]^{4-}$ based on crystal field theory is
 (1) $t_{2g}^4 e_g^2$ (2) $t_{2g}^6 e_g^0$
 (3) $t_{2g}^3 e_g^3$ (4) $t_{2g}^2 e_g^4$

Space for Rough Work

BOTANY**SECTION - A**

101. The basis of DNA fingerprinting is
- (1) Variability in size of genome
 - (2) The sequence of coding and non-coding regions in the chromosome
 - (3) Variable number of tandem repeats
 - (4) Complementarity of DNA
102. Which among the following statements is **not** true for human genome?
- (1) Repeated sequences make up very large portion of the human genome
 - (2) Chromosome X has most number of genes
 - (3) The total number of genes is estimated at 30000
 - (4) Less than 2% of the genome codes for proteins
103. Select the **mismatched** pair w.r.t. *lac* operon.
- | | | | |
|-----|-----------------|---|--|
| (1) | Operator gene | – | Interacts with a regulator molecule |
| (2) | Promoter gene | – | Possess site for RNA polymerase binding |
| (3) | Structural gene | – | Transcribe mRNA for polypeptide synthesis |
| (4) | Regulatory gene | – | Functional in the presence of lactose only |
104. Which of the following codons has two functions?
- (1) UUA
 - (2) UGA
 - (3) AUG
 - (4) UAG
105. George Gamow suggested that
- (1) A single codon can code for more than one amino acid
 - (2) A genetic code should constitute a combination of bases
 - (3) The codon in mRNA should be read in contiguous fashion
 - (4) The information of the amino acid sequence in proteins is present in DNA

106. Which among the following is **not** catalysed by RNA polymerase III?

- (1) 5S rRNA
- (2) ScRNA
- (3) tRNA
- (4) hnRNA

107. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	tRNA	(i)	Perform catalytic function during translation
b.	rRNA	(ii)	Provides template for translation
c.	hnRNA	(iii)	Brings amino acid during translation
d.	mRNA	(iv)	Precursor of messenger RNA

- | | a | b | c | d |
|-----|-------|-------|-------|------|
| (1) | (ii) | (iii) | (i) | (iv) |
| (2) | (iii) | (i) | (iv) | (ii) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (ii) | (i) | (iii) | (iv) |

108. How many types of DNA dependent RNA polymerase catalyses the transcription of mRNA, tRNA and rRNA in bacteria?

- (1) Two
- (2) Three
- (3) Single
- (4) Five

109. In post transcriptional processing, splicing is

- (1) Joining of two mRNAs
- (2) Removal of exons
- (3) Fragmentation of hnRNA
- (4) Removal of introns and joining of exons in defined order

110. Read the following statements and select the **correct** option.

Assertion (A): Replication of DNA occurs within a small opening of the DNA helix, referred to as replication fork.

Reason (R): Due to very high energy requirements the two strands of DNA cannot be separated in its entire length.

Space for Rough Work

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) Both (A) and (R) are false.
111. Select the **incorrect** statement from the following.
 (1) DNA acts as genetic material in eukaryotes and prokaryotes
 (2) Nucleic acids do not function as structural molecules
 (3) RNA can also act as genetic material
 (4) RNA mostly functions as a messenger
112. The term 'Nuclein' is referred to
 (1) Semifluid matrix present in the nucleus
 (2) Fluid present in perinuclear space
 (3) Components of nucleolus
 (4) An acidic substance present in nucleus
113. Genetic material in which of the following has least number of nucleotides?
 (1) *Escherichia coli*
 (2) $\phi \times 174$ bacteriophage
 (3) Bacteriophage λ
 (4) Human WBC
114. A bacterial cell has 2.42×10^6 bases in its circular DNA. The number of phosphodiester bonds present in this DNA would be
 (1) 4.82×10^6 (2) 2.42×10^{12}
 (3) 2.42×10^6 (4) 1.21×10^6
115. Select the nitrogenous base(s) which is/are **not** common between DNA and RNA.
 a. Thymine b. Guanine
 c. Adenine d. Uracil
 e. Cytosine
 (1) d only (2) b, c and e
 (3) a and d (4) a only
116. In DNA, phosphodiester linkage is found between
 (1) 3' carbon and 5' carbon of two different nucleotides
 (2) 3' carbon of two different nucleotides
 (3) 5' carbon of a deoxyribose sugar and 7' carbon of a purine base
 (4) 3' carbon of a deoxyribose sugar and 5' carbon of ribose sugar
117. Which of the following confer(s) stability to the helical structure of DNA?
 a. Presence of 9-membered double ring structure of purines.
 b. Stacking of planes of base pair one over the other.
 c. Occurrence of hydrogen bonds between the bases of each base pair.
 The **correct** one(s) is/are
 (1) c only (2) b only
 (3) a and b (4) b and c
118. All of the following are true w.r.t. dsDNA, **except**
 (1) Purines are generally more in number as compared to pyrimidines
 (2) Adenine is equimolar with thymine
 (3) Sugar deoxyribose and phosphate residues occur in equal number
 (4) The ratio (adenine + guanine) : (thymine + cytosine) is constant for all species
119. Read the following statements.
Statement A: The process by which characters are passed on from parent to progeny is called inheritance.
Statement B: Inheritance is the basis of heredity.
 In the light of above statements choose the **correct** option.
 (1) Only statement A is correct
 (2) Only statement B is correct
 (3) Both statements A and B are correct
 (4) Both statements A and B are incorrect
120. Match the terms given in Column-I with their appropriate meaning in Column-II and select the most suitable option.
- | Column-I | Column-II |
|-------------------------|--|
| a. Alleles | (i) F_1 individual resembles only one of the parents |
| b. Co-dominance | (ii) F_1 phenotype is in-between the parents |
| c. Incomplete dominance | (iii) F_1 resembles both the parents |
| d. Dominance | (iv) Slightly different forms of the same gene |
- (1) a(iv), b(iii), c(ii), d(i) (2) a(iii), b(ii), c(i), d(iv)
 (3) a(iv), b(i), c(ii), d(iii) (4) a(iii), b(i), c(iv), d(ii)

Space for Rough Work

121. Which among the following Mendelian crosses will result in 9 : 3 : 3 : 1 phenotypic ratio in next generation in pea plant?

- (1) $RRyy \times rrYy$ (2) $RRYy \times Rryy$
(3) $RrYy \times RrYy$ (4) $RRyy \times rrYY$

122. How many different types of gametes will be formed by a heterozygous individual having genotype $AaBbCc$?

- (1) 2^2 (2) $3 \times 3 \times 3$
(3) 6 (4) 2^3

123. What would be the ratio of different genotypes to different phenotypes regarding ABO blood grouping in human population?

- (1) 1 : 2 (2) 3 : 2
(3) 7 : 4 (4) 2 : 3

124. Who used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and mapped their position on the chromosome?

- (1) Theodore Boveri (2) Walter Sutton
(3) T.H. Morgan (4) Alfred Sturtevant

125. Read the following statements and select the correct option.

Assertion (A): Phenylketonuria affected individual has the accumulation of phenylpyruvic acid and other derivatives of phenylalanine in the brain.

Reason (R): Phenylketonuria affected individual has an enzyme that converts the phenylalanine into tyrosine.

In the light of the above statements, choose the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

126. If mother and father both are carrier for the gene, i.e. heterozygous for sickle cell anaemia, then the chances of offsprings being

- (1) Heterozygous for the genes is 75%
(2) Diseased is 50%
(3) A carrier is 25%
(4) Homozygous for sickle cell anaemia is 25%

127. Select the **mismatched** pair.

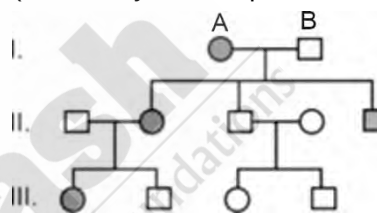
- (1) Klinefelter's syndrome – Gynaecomastia
(2) Turner's syndrome – Rudimentary ovaries
(3) Haemophilia – Extremely rare in females
(4) Down's syndrome – Observed only in males

128. If size of starch grains in pea is considered as phenotype, the heterozygous condition of alleles (Bb) will show

- (1) Dominant phenotype
(2) Incomplete dominance
(3) Co-dominance
(4) Recessive phenotype

129. A pedigree chart for a Mendelian character is given below.

(Shaded symbols represent affected individuals)



Find the possible genotypes for A and B:

- (1) ♀(Aa) and ♂(aa)
(2) ♀(Aa) and ♂(Aa)
(3) ♀($Hb^A Hb^A$) and ♂($Hb^S Hb^S$)
(4) ♀(aa) and ♂(AA)

130. In *Antirrhinum*, a pink flower is crossed with white flower. What proportion of red flowers will be produced in F_1 generation?

- (1) 0% (2) 25%
(3) 75% (4) 50%

Space for Rough Work

131. Select the **incorrect** statement from the following.
- (1) To determine the genotype of a tall pea plant in F_2 generation, Mendel performed a test-cross
 - (2) Law of segregation is based on the fact that the alleles do not show any blending and both characters are recovered as such in F_2 -generation
 - (3) AB blood group in humans is an example of incomplete dominance
 - (4) Multiple alleles can be formed only when population studies are made
132. The graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross, from a diagram was developed by
- (1) Reginald C. Punnett
 - (2) Tschermak
 - (3) Hugo de Vries
 - (4) Correns
133. How many true-breeding pea plant varieties were selected by Mendel for his experiments, as pairs which were similar except for one character with contrasting traits?
- (1) Seven
 - (2) Twelve
 - (3) Fourteen
 - (4) Twenty
134. Consider the following statements regarding genes and chromosomes.
- a. Sutton and Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes.
 - b. The two alleles of a gene pair are located on homologous sites of non-homologous chromosomes.
 - c. Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it chromosomal theory of inheritance.
 - d. Experimental verification of chromosomal theory of inheritance was done by T.H. Morgan.
- Which of the above statements are **correct**?
- (1) All a, b, c and d
 - (2) a, c and d only
 - (3) a and b only
 - (4) b, c and d only
135. A couple is having a son and a daughter. What is the probability, that their third child will be a daughter again?
- (1) 25%
 - (2) 50%
 - (3) 75%
 - (4) 0%

SECTION - B

136. How many among the following are dominant traits of pea plant?
- Green pod colour, constricted pod shape, yellow seed colour, wrinkled seed shape, green seed colour, violet flower colour.
- (1) Two
 - (2) Three
 - (3) Four
 - (4) Five
137. Which among the following does **not** show quantitative inheritance?
- (1) Kernel colour in wheat
 - (2) Height in humans
 - (3) Skin colour in humans
 - (4) Plant height in pea plant
138. Which among the following genotypes of organism is showing monohybrid condition?
- (1) RRppQQ
 - (2) RRppqq
 - (3) RRPpQQ
 - (4) RrPpQq
139. Read the following statements and choose the **correct** option.
- Statement (A):** On a chromosome when genes were tightly linked, they will show very low recombination frequency.
- Statement (B):** The dihybrid cross on *Drosophila* conducted by Morgan shows higher non-parental type gene combination than the parental type gene combination.
- (1) Only statement A is correct
 - (2) Only statement B is correct
 - (3) Both statements A and B are correct
 - (4) Both statements A and B are incorrect
140. Sickle cell anaemia is caused by the substitution of A by B at the sixth position of the beta globin chain of the haemoglobin molecule. Choose the correct option for A and B.

	A	B
(1)	Glutamine	Valine
(2)	Valine	Glutamic acid
(3)	Glutamic acid	Valine
(4)	Valine	Glutamine

Space for Rough Work

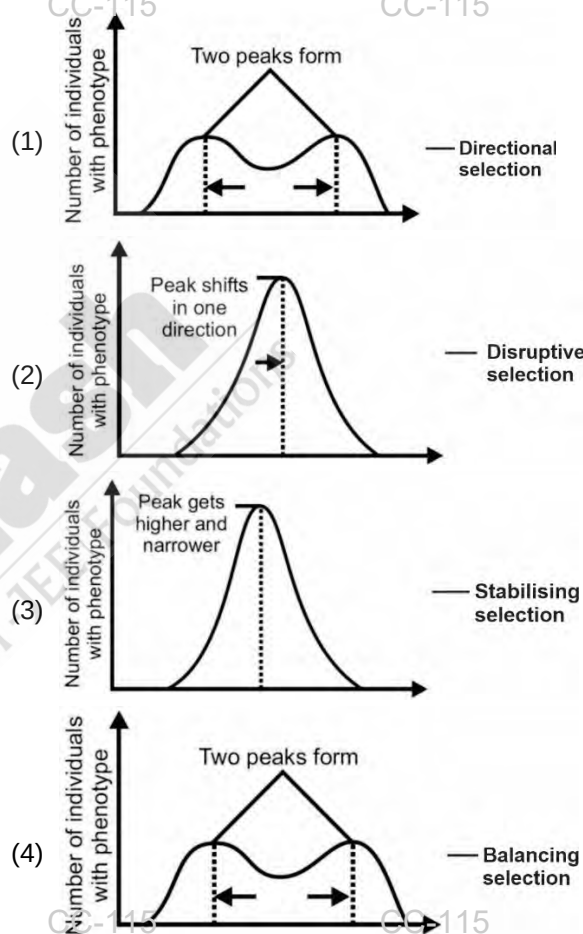
141. XO type of sex-determination in which the males have only one X-chromosome besides the autosomes, whereas females have a pair of X-chromosomes besides autosomes is seen in
 (1) Grasshopper (2) *Drosophila*
 (3) Butterflies (4) Birds
142. Select the statement which is **odd** one out w.r.t. polygenic inheritance.
 (1) The phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive
 (2) Besides involvement of multiple genes it also takes into account the influence of environment
 (3) Polygenic traits show distinct alternate forms in their occurrence and they are not spread across a gradient
 (4) Human skin colour is a classic example of polygenic inheritance
143. The haplo-diploid sex-determination system, as seen in honeybee, has characteristic features such as
 (1) Males produce sperms by mitosis
 (2) Worker bees are developed from unfertilised eggs
 (3) Males have fathers and thus can have sons
 (4) Males do not have a grandfather and so cannot have grandsons
144. Regarding central dogma of molecular biology which statement given below is **incorrect**?
 (1) It explains unidirectional flow of information
 (2) Information from DNA flows to RNA
 (3) Amino acid sequence in a polypeptide codes the information in mRNA
 (4) Information flows from RNA to trait expressing molecule
145. In the transformation experiment conducted by Griffith, when the mixture of heat killed S strain *Pneumococcus* and live R strain *Pneumococcus* was injected to mice, the mice died. What should be the most appropriate reason behind this?
 (1) R strain bacteria were transformed into S strain
 (2) S strain bacteria became alive
 (3) S strain bacteria were transformed in R strain
 (4) DNA of died S strain bacteria started to synthesize toxic proteins
146. A molecule that acts as genetic material should provide the scope for slow mutation. This is required for
 (1) Quick duplication of genetic material
 (2) Evolution of life forms
 (3) Genetic recombination during sexual reproduction
 (4) Replication with highest degree of accuracy
147. Ribozymes are
 (1) Protein catalysts present in ribosomes
 (2) Enzymes that degrade RNA polymers
 (3) RNA molecules that act as catalyst
 (4) Enzymes that dissociate the sub-units of a ribosome from each other
148. An *E. coli* cell has hybrid DNA with one strand of ^{14}N and other of ^{15}N . If this cell is allowed to grow in a medium containing $^{14}\text{NH}_4\text{Cl}$ as the only nitrogen source, then what percent of cells in the third generation will have hybrid DNA?
 (1) 25% (2) 50%
 (3) 12.5% (4) 87.5%
149. Function of DNA ligase is to catalyze the
 (1) Formation of glycosidic bond between nitrogenous base and pentose sugar
 (2) Joining of two okazaki fragments during replication
 (3) Breakage of hydrogen bonds between two strands
 (4) DNA polymerisation in $3' \rightarrow 5'$ direction
150. The DNA sequence that provides binding site for RNA polymerase is present in
 (1) Structural gene (2) Terminator region
 (3) Coding strand only (4) Promoter region

Space for Rough Work

ZOOLOGY

SECTION - A

151. S.L. Miller created electric discharge in a closed flask containing methane, ammonia, hydrogen and water vapour at
- (1) 150°C (2) 100°C
(3) 80°C (4) 800°C
152. On Galapagos islands, _____ were the ancestral finches from which many other finches evolved. Choose the option which fills the blank **correctly**.
- (1) Insect-eating finches
(2) Fruit-eating finches
(3) Cactus-eating finches
(4) Seed-eating finches
153. Around 200 mya, some of the land reptiles went back into water and evolved into
- (1) Fish-like reptiles
(2) Mammal-like amphibians
(3) Fish-like amphibians
(4) Reptile-like fishes
154. Which of the following is used as a male contraceptive?
- (1) Diaphragm (2) Nirodh
(3) Cervical cap (4) Multiload 375
155. _____ was about 20 feet in height and had huge fearsome dagger-like teeth. Select the **correct** option to fill in the blank.
- (1) *Stegosaurus* (2) *Triceratops*
(3) *Tyrannosaurus* (4) *Brachiosaurus*
156. All of the following are natural contraceptive methods, **except**
- (1) Withdrawal method
(2) Coitus interruptus
(3) Tubectomy
(4) Lactational amenorrhoea
157. Consider a population which is in Hardy-Weinberg equilibrium where 64% of people show homozygous dominant trait. What is the percentage of people which are heterozygous for the same character?
- (1) 36% (2) 32%
(3) 16% (4) 84%
158. Select the option representing **correct** match w.r.t. diagrammatic representation of the operation of natural selection on different traits with types of natural selection.



Space for Rough Work

159. All of the following are extinct vertebrates, **except**
- Pelycosaurs
 - Tuataras
 - Thecodonts
 - Therapsids
160. A fossil of 'X' was discovered in Java in 1891 and had a large brain size of around 900cc. Identify 'X' and select the correct option.
- Dryopithecus*
 - Neanderthal man
 - Homo erectus*
 - Homo habilis*
161. Complete the analogy.
Lemur : Spotted cuscus :: Anteater : _____
Choose the **correct** option.
- Marsupial mole
 - Bobcat
 - Numbat
 - Tasmanian tiger cat
162. Select the option which represents **correct** expansion of the following abbreviations.
- ZIFT – Zygote Intra Fallopian Transfer
 - ICSI – Inter Cytoplasmic Semen Injection
 - GIFT – Gamete Intra Fallopian Technology
 - IUT – Inter Uterine Transfer
163. According to modern view, evolution can be described as a result of processes which are based on chance events in nature and chance mutations in organisms. Nature of these processes are
- Stochastic
 - Non-directional
 - Non-random
 - Never anthropogenic
- Choose the **correct** option.
- (a) and (b) only
 - (b) and (c) only
 - (a), (c) and (d)
 - (a) and (d)
164. Which of the following is **true** for vasectomy?
- Sperm production decreases
 - Degeneration of testes
 - Loss of sexual desire
 - Blocks gamete transport and thereby prevents conception
165. The primates that used hides to protect their body and buried their dead were
- Dryopithecus*
 - Ramapithecus*
 - Australopithecines*
 - Neanderthal men
166. Read the following features.
- Cover the cervix during coitus
 - Block entry of sperms through the cervix
 - Their contraceptive efficiency increases when used along with spermicidal creams
 - They are reusable
- The above mentioned features are applicable for which of the following contraceptives?
- Condoms
 - Vaults
 - Lippes loop
 - Progestasert
167. Suppose a female is completely healthy for conceiving but the male partner is suffering from erectile dysfunction and is unable to inseminate the female, then the best suitable ART that should be opted by that couple for having a baby is
- GIFT
 - ICSI
 - AI
 - ZIFT
168. The zygote or early embryo upto 8 blastomeres is transferred into the A and this is called B. Identify A and B and select the correct option.
- | A | B |
|--------------------|------|
| (1) Fallopian tube | GIFT |
| (2) Uterus | IUI |
| (3) Uterus | IUT |
| (4) Fallopian tube | ZIFT |
169. All of the following are grounds for MTP in India, **except**
- Continuation of pregnancy can cause harm to foetus
 - Pregnancy is a result of casual unprotected sexual intercourse
 - Pregnancy is due to failure of contraceptive
 - Abortion after finding the sex of foetus to be a female
170. The 'Theory of special creation' has few connotations. Select the odd one out w.r.t. this theory.
- All living organisms that we see today were created as such.
 - All the existing life forms share similarities and share common ancestors
 - Earth is about 4000 years old.
 - The diversity was always the same since creation and will be the same in future also.

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171. Sphenopsids (Horsetails) were directly evolved from
 (1) Tracheophyte ancestors
 (2) Rhynia-type plants
 (3) Chlorophyte ancestors
 (4) Psilophyton
172. Vertebrate hearts or brains show
 a. Divergent evolution b. Convergent evolution
 c. Homology
 Select the correct option.
 (1) a and b only (2) b and c only
 (3) a, b and c (4) a and c only
173. Which of the following animals was a flying reptile?
 (1) *Triceratops* (2) *Ichthyosaurs*
 (3) *Stegosaurus* (4) *Pteranodon*
174. After industrialisation, the gene frequency of white-winged moths decreased. This is because of
 (1) Change in genetic make up of white-winged moth due to smoke
 (2) Change in behaviour of predator of moth due to smoke
 (3) Deforestation due to anthropogenic effect
 (4) Easy detection of white-winged moths by predators against a contrasting background
175. The Big Bang theory talks of a singular huge explosion, unimaginable in physical terms. This theory attempts to explain to us the origin of
 (1) Planets (2) Solar system
 (3) Universe (4) Milky way galaxy
176. Match column I with column II and select the correct option w.r.t. evolution of respective primates.

	Column I		Column II
a.	<i>Dryopithecus</i>	(i)	1.5 mya
b.	<i>Australopithecines</i>	(ii)	2 mya
c.	<i>Homo erectus</i>	(iii)	15 mya
d.	<i>Homo sapiens</i>	(iv)	75,000-10,000 years ago

- | | | | | |
|-----|----------|----------|----------|----------|
| | a | b | c | d |
| (1) | (iii) | (i) | (ii) | (iv) |
| (2) | (ii) | (i) | (iii) | (iv) |
| (3) | (iii) | (ii) | (i) | (iv) |
| (4) | (iv) | (i) | (ii) | (iii) |

177. Which of the following statements is incorrect?
 (1) About 2000 mya, the first cellular forms of life appeared on Earth.
 (2) About 65 mya, the dinosaurs suddenly disappeared from the Earth.
 (3) Due to the continental drift, pouched mammals of Australia survived because of lack of competition from any other mammal.
 (4) By the time of 500 mya, vertebrates were formed and active.
178. In 1938, a fish caught in _____ happened to be a _____ which was thought to be extinct. These animals called lobefins evolved into the first amphibians that lived on both land and water. Select the correct option to fill in the blanks respectively.
 (1) South America, *Ichthyophis*
 (2) South Africa, *Coelacanth*
 (3) South Africa, *Ichthyosaur*
 (4) North Africa, *Coelacanth*
179. Consider the following statements and choose the correct option.
Statement A : Darwinian variations are small and directional while de Vries mutations are random and directionless.
Statement B : If the variation in gene frequencies within a population occurs by chance, it is called gene flow.
 (1) Both statements A and B are correct
 (2) Both statements A and B are incorrect
 (3) Only statement A is correct
 (4) Only statement B is correct
180. Which of the following statements is correct?
 (1) Saltation is a directional process which takes place in multiple steps
 (2) Founder effect is a ramification of genetic drift
 (3) Darwin's finches represent one of the best examples of convergent evolution
 (4) The phenomenon of evolution of different species in a given geographical area starting from a point and spreading to other habitats is called co-evolution.

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181. Theory of panspermia is related to
 (1) Chemical evolution (2) Origin of Earth
 (3) Origin of life (4) Abiogenesis
182. Which of the following options is correct w.r.t. evolution of man?
 (1) *Ramapithecus* was more ape-like.
 (2) *Australopithecines* probably lived in East Asian grasslands.
 (3) *Homo sapiens* arose in Africa and moved across continents and developed into distinct races.
 (4) *Homo habilis* probably ate meat
183. Select the correct set of STIs which can be completely curable, if detected early and treated properly.
 (1) Gonorrhoea, Genital herpes, AIDS
 (2) Hepatitis-B, Genital warts, AIDS
 (3) Gonorrhoea, Genital warts, Trichomoniasis
 (4) AIDS, Genital Herpes, Genital warts
184. Choose the incorrect match.
- | | | | |
|-----|---------------|---|---------------------------------|
| (1) | Multiload 375 | – | Hormone releasing IUD |
| (2) | Lippes loop | – | Non-medicated IUD |
| (3) | Diaphragm | – | Barrier method of contraception |
| (4) | Progestasert | – | Medicated IUD |
185. Which of the following can be a complication resulting from STIs?
 (A) PID (B) Still birth
 (C) Ectopic pregnancies (D) Infertility
 Select the correct option.
 (1) A, B and D only (2) C and D only
 (3) A, B, C and D (4) A and B only

SECTION - B

186. Which of the following events occurred first during the course of origin of life?
 (1) Formation of protobionts
 (2) Synthesis of organic monomers
 (3) Synthesis of organic polymers
 (4) Formation of DNA-based genetic systems
187. Which of the following is not a venereal disease?
 (1) Chlamydiasis (2) Trichomoniasis
 (3) Diphtheria (4) Syphilis

188. How many of the following statements is/are correct w.r.t. origin of life?
 (a) The earliest organisms that appeared on the Earth were non-green and presumably anaerobes.
 (b) The first autotrophic organisms were the chemoautotrophs that never released oxygen.
 (c) All life forms were in water environment only.
 (d) The first non-cellular forms of life could have originated 3 billion years back.

Select the correct option.

- (1) Four (2) Three
 (3) Two (4) One
189. The study of history of life forms on earth is called
 (1) Biodiversity (2) Evolutionary biology
 (3) Paleontology (4) Embryology
190. Select the **odd one** w.r.t. mammals who live wholly in water.
 (1) Platypus (2) Sea cows
 (3) Whales (4) Dolphins
191. Read the following statements A and B and choose the correct option.
Statement A : Similarities in proteins and genes performing a given function among diverse organisms gives clues to common ancestry.
Statement B : Animals with common ancestry show convergent evolution.
 (1) Both statements A and B are correct
 (2) Both statements A and B are incorrect
 (3) Only statement A is incorrect
 (4) Only statement B is incorrect
192. Choose the correct match.

(1)	Human settlement started	–	1,00,000 years back
(2)	Origin of Earth	–	4.5 million years ago
(3)	Origin of Universe	–	20 million years ago
(4)	Existence of sea weeds and few plants	–	320 million years ago

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193. **Assertion (A):** Periodic abstinence is a method of conception exercised during 10th - 17th day of the 32 days menstrual cycle by a normal healthy couple.
Reason (R): Minimum chances of conception are possible during the duration of 10th to 17th day of the 32 days menstrual cycle.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 (3) (A) is correct but (R) is incorrect.
 (4) Both (A) and (R) are incorrect.
194. How many of the contraceptives given in the box below make uterus unsuitable for implantation and alter the quality of cervical mucus?
- | |
|---|
| LNG-20, Progestasert, Saheli, OCPs, Implants, Multiload 375 |
|---|
- Select the correct option.
 (1) Three (2) Four
 (3) Six (4) Five
195. Read the following statements.
Statement A : In India, there is statutory ban on amniocentesis for sex determination to legally check increasing menace of female foeticides.
Statement B : In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells and dissolved substances.
 Select the correct option.
 (1) Both statements A and B are correct
 (2) Both statements A and B are incorrect
 (3) Only statement A is correct
 (4) Only statement B is correct
196. A diabetic patient with 22 weeks of pregnancy wants to terminate her pregnancy because of her health issues. Opinion of how many registered medical practitioner(s) is/are required in above case under MTP (Amendment) Act, 2017?
 (1) One (2) Three
 (3) Two (4) Four
197. All of the following are incorrect w.r.t. 'Saheli', **except**
 (1) It is only a post-coital contraceptive pill.
 (2) Blocks estrogen receptors in the uterus, preventing embryo from getting implanted
 (3) Steroidal preparation, either progesterone or combination of progesterone or estrogen
 (4) Protects the endometrial lining from adverse effect of estrogen
198. Role of copper ions in CuT is to
 (1) Suppress sperm motility
 (2) Decrease phagocytosis of sperm
 (3) Stimulate ovulation
 (4) Make uterus suitable for implantation
199. Which of the following is not the probable reason for rapid increase in population growth?
 (1) Rapid decline in death rate
 (2) Decline in MMR
 (3) Decline in IMR
 (4) Decrease in number of people in reproductive age
200. Test tube baby programme is
 (1) *In-vitro* fertilization followed by the transfer of embryo into the fallopian tube or uterus
 (2) *In-vivo* fertilization in a female followed by implantation of embryo in her reproductive tract
 (3) *In-vitro* fertilization followed by implantation of zygote into the uterus
 (4) Fertilization inside the body followed by embryo transfer in test tube



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